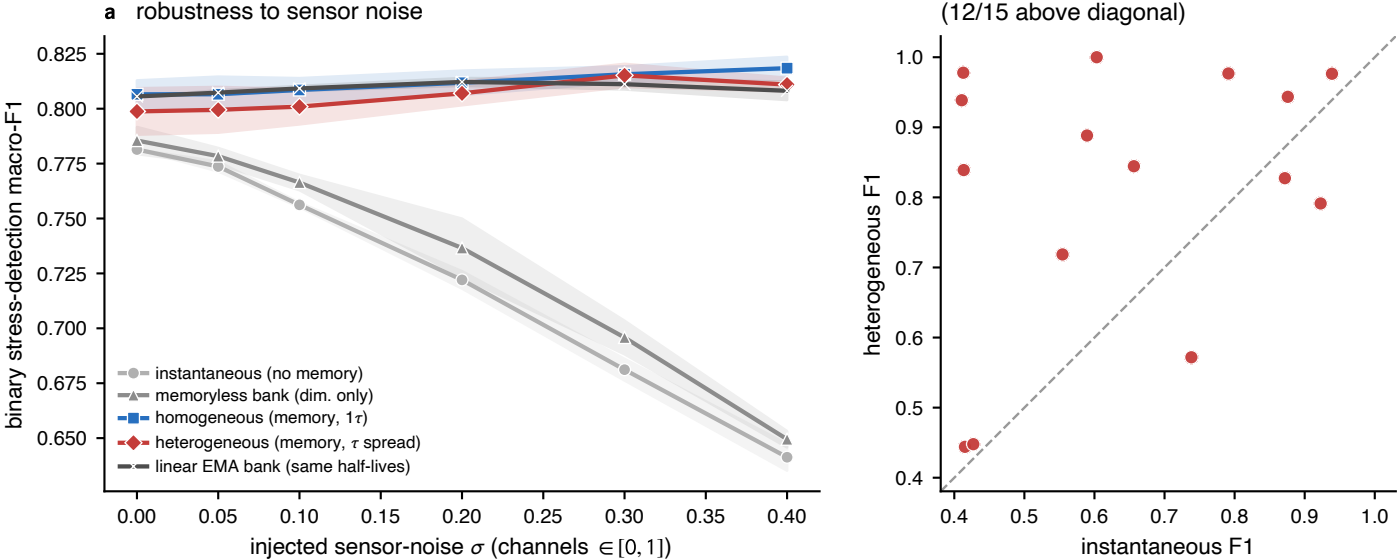
a robustness to sensor noise
binary stress-detection macro-F1
0.825
0.800
0.775
0.750
0.725
0.700
0.675
0.650
instantaneous (no memory)
memoryless bank (dim. only)
homogeneous (memory, 1τ)
heterogeneous (memory, τ spread)
linear EMA bank (same half-lives)
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
injected sensor-noise σ (channels ∈ [0, 1])
b per subject, σ = 0.4
(12/15 above diagonal)
heterogeneous F1
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.4
0.5
0.6
0.7
0.8
0.9
1.0
instantaneous F1